

MOHAMMED MAHYOUB

Senior System Integration & ITS Engineer | AI & Computer Vision

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PROFESSIONAL SUMMARY

Senior System Integration and ITS Engineer with 14+ years of multidisciplinary engineering experience, including 6+ years delivering smart infrastructure and intelligent traffic enforcement systems across the UAE. Experienced in integrating cameras, LiDAR, sensors, networks, control units, databases, and backend platforms into reliable end-to-end operational systems. Currently pursuing an MSc in Artificial Intelligence at UWE Bristol, with a focus on machine learning, computer vision, smart mobility, and road safety.

CORE EXPERTISE

System Integration | ITS & Traffic Enforcement | Smart Infrastructure | FAT / SAT | Commissioning & Calibration | Machine Vision & LiDAR | Networking & Backend Integration | Root-Cause Analysis | Systems Engineering | Python & Machine Learning

PROFESSIONAL EXPERIENCE

Senior Service Engineer – Smart Infrastructure & ITS Systems | VITRONIC Machine Vision

Oct 2019 – Present | United Arab Emirates

- Deploy, integrate, commission, and support intelligent traffic enforcement and ITS solutions across mission-critical sites in the UAE.
- Integrate cameras, LiDAR, sensors, control units, networking infrastructure, databases, and backend platforms into reliable end-to-end operational systems.
- Execute FAT, SAT, commissioning, calibration, interface validation, acceptance testing, and performance verification activities.
- Diagnose complex hardware, software, communication, database, and system-interface failures using structured root-cause analysis.
- Coordinate incident response, preventive maintenance, upgrades, configuration changes, and technical resolution with customers, vendors, and multidisciplinary teams.
- Prepare technical reports, test documentation, incident analyses, and system handover records while supporting high availability and service continuity.

Electronics, Embedded Systems & Automation Engineer | Electronic Science Company

Jun 2015 – Aug 2019 | Al Hudaydah, Yemen

- Designed, assembled, tested, and troubleshoot electronic circuits, embedded control systems, and sensor-based automation solutions.
- Programmed PIC, AVR, and Arduino microcontrollers and integrated sensors, displays, communication modules, controllers, and peripherals.
- Performed PCB prototyping, hardware debugging, functional testing, and technical validation.
- Developed VB.NET desktop applications for monitoring, diagnostics, reporting, and workflow automation, and provided deployment and customer support.

Independent Engineering Consultant – Network Infrastructure & Technical Projects | Independent Engineering Projects

Mar 2012 – May 2018 | Independent / Project-based

- Delivered selected independent engineering projects alongside professional employment across network infrastructure, wireless systems, automation, simulation, and renewable-energy applications.

- Designed and supported routing, switching, firewall, VPN, MikroTik, and Ubiquiti environments, including wireless coverage and point-to-point link planning.
- Designed and simulated MPLS, OSPF, BGP, VRF, and traffic-engineering architectures using GNS3 and Cisco platforms.
- Prepared network diagrams, configurations, RF calculations, engineering studies, simulations, and technical reports.

EDUCATION

Master of Science (MSc), Artificial Intelligence | University of the West of England (UWE Bristol)

May 2026 – Jun 2028 | In progress

Focus: machine learning, neural networks, big data, AI optimisation, computer vision, and intelligent system integration.

Bachelor's Degree, Electrical, Electronics & Communications Engineering | Sana'a University

Sep 2012 - Jun 2017 | UAE Ministry of Education equivalency obtained

CERTIFICATIONS

Certificate of Appreciation - Dubai Police | Dubai Police HQ, UAE | 2023

CompTIA A+ | CompTIA

Cisco Certified Network Associate Routing and Switching (CCNA) | Cisco

SELECTED PROJECTS

- Vehicle Type Classification Using NGSIM US-101: preprocessing, EDA, feature engineering, Logistic Regression, Decision Tree, Random Forest, GridSearchCV, and leakage-resistant vehicle-level evaluation.
- Embedded Systems, IoT & Industrial Automation: microcontrollers, PLC platforms, sensors, communication modules, UAV electronics, and real-time monitoring concepts.
- Wireless & RF Network Infrastructure: multi-zone LAN/WLAN architecture, MikroTik and Ubiquiti design, point-to-point links, VPN, surveillance, and access-control connectivity.

TECHNOLOGIES & LANGUAGES

Technologies: Python, scikit-learn, VB.NET, Arduino, PIC, AVR, Cisco, MikroTik, Ubiquiti, GNS3, VMware ESXi, FreeNAS/iSCSI, Citrix XenApp, cameras, LiDAR, sensors, databases, and backend platforms.

Languages: Arabic (Native or Bilingual) | English (Professional Working Proficiency)